



CAD Trust v2.0.2 — Machine-Readable Schema Reference

This page is the developer reference for the CAD Trust Data Model v2.0.2 machine-readable schema. It covers every table, every field, all picklist values, validation rules, and migration notes from v1.

Schema file: `cadtrust-schema-v2.0.2.json` · Version `2.0.2` · Published `27 January 2026`

Source: [Data Dictionary v2.0.2 \(Excel\)](#)

ERD: [Live Entity Relationship Diagram](#)

What This Schema Is For

Registries connecting to CAD Trust v2.0 need to map their internal data to the CAD Trust data model before they can broadcast to the network. This schema makes that process machine-assisted rather than manual.

Use it to:

- **Drive ETL mapping tools** — load the schema, iterate over tables and fields, auto-suggest registry field → CAD Trust field mappings
- **Validate data before submission** — check required fields, enum values, date formats, and foreign key consistency without running a full Chia node
- **Build a sandbox validator** — accept registry data via API, run validation against this schema, return structured error reports
- **Generate documentation** — auto-render field reference pages, API docs, and integration guides from the schema
- **Power the AI mapping tool** — feed as context to an LLM to suggest mappings from a registry's own schema

Schema Structure

The JSON file has six top-level sections:

Section	Purpose
<code>\$schema</code> , <code>\$id</code> , <code>title</code> , <code>version</code>	Standard JSON Schema metadata
<code>changelog</code>	What changed from v1 to v2.0 — new tables, restructured tables, deprecated fields
<code>picklists</code>	All allowed enum values for every picklist field, by field name
<code>tables</code>	Full field definitions for all 15 tables
<code>validationRules</code>	Required fields for data activation, conditional requirements, enum notes
<code>migrationNotes</code>	Step-by-step guidance for v1 → v2 migration per changed concept

Quick Start — Reading the Schema

```
import json
import requests

# Load schema
with open('cadtrust-schema-v2.0.2.json') as f:
    schema = json.load(f)
```

```
# List all tables
print(list(schema['tables'].keys()))
# ['project', 'methodology', 'project_methodology', 'location',
#  'validation', 'verification', 'issuance', 'unit', 'stakeholder',
#  'label', 'unit_label', 'co_benefit', 'estimation', 'rating', 'program']

# Get all required fields for the project table
project_required = [
    field for field, meta
    in schema['tables']['project']['fields'].items()
    if meta['required'] == True
]
print(project_required)

# Get all allowed values for unit_status
print(schema['picklists']['unit_status'])
# ['Issued', 'Held', 'Retired', 'Cancelled', 'Expired', 'Buffer', 'Inactive', 'Rejected']

# Get the full list of fields required for data activation
print(schema['validationRules']['required_for_data_activation'])
```

```
// JavaScript / Node.js
const schema = require('./cadtrust-schema-v2.0.2.json');

// Validate a field value against a picklist
function validatePicklist(fieldName, value) {
    const allowed = schema.picklists[fieldName];
    if (!allowed) return { valid: true }; // no picklist constraint
    return {
        valid: allowed.includes(value),
        allowed,
        error: allowed.includes(value) ? null : `'${value}' not in allowed values for ${fieldName}`
    };
}

console.log(validatePicklist('unit_status', 'Active'));
// { valid: false, error: "'Active' not in allowed values for unit_status" }
// NOTE: 'Active' was renamed to 'Held' in v2.0

console.log(validatePicklist('unit_status', 'Held'));
// { valid: true }
```

Tables Overview

15 tables total. 6 are new in v2.0. 2 were restructured.

Table	Status	Description	Parent FK
project	Existing	Core project record. All project-level tables reference this.	—
methodology	NEW v2.0	Standalone methodology registry with versioning	—
project_methodology	NEW v2.0	Junction: project ↔ methodology (many-to-many)	project, methodology
location	Enhanced	Project geographic location. New: location_map_type , location_map_file_link	project
validation	Restructured	Validation event(s) for a project. Now separate from verification.	project

Table	Status	Description	Parent FK
verification	Restructured	Periodic verification events. Links to validation. Referenced by issuance.	project, validation
issuance	Enhanced	Unit issuance event. Now requires FK to project_methodology .	verification, project_methodology, location
unit	Enhanced	Credit unit block. New: retirement detail, beneficiary, LEI, ITMOs reference ID.	issuance
stakeholder	NEW v2.0	Named project participants (owner, developer, consultant)	project
label	NEW v2.0	Credit eligibility labels (CORSIA, CCP, ITMO, etc.)	—
unit_label	NEW v2.0	Junction: unit ↔ label (many-to-many)	unit, label
co_benefit	Existing	SDG co-benefits linked to a project	project
estimation	Existing	Expected credit volumes for a future crediting period	project
rating	Existing	Independent project ratings (BeZero, CDP, CCQI, etc.)	project
program	NEW v2.0	Programme or activity grouping. Referenced from project.	—

AEF tables (`aef_t1_submission` , `aef_t2_authorizations` , `aef_t3_actions` , `aef_t4_holdings` , `aef_t5_authorized_entities`) are defined in the full schema JSON. They support Article 6 UNFCCC reporting and are only required for registries submitting AEF reports. See the schema JSON for full field definitions.

Field Definition Format

Every field in `tables.<table>.fields` follows this structure:

```
"unit_vintage_year": {
  "column": "unit_vintage_year",
  "apiField": "unitVintageYear",
  "type": "Year",
  "required": true,
  "source": "User Input",
  "description": "Year the units were awarded (vintage).",
  "aefMapping": "Vintage",
  "example": "2023"
}
```

Property	Meaning
column	Database column name (snake_case) — use in SQL/ETL
apiField	API field name (camelCase) — use in JSON payloads
type	Data type: <code>String</code> , <code>Text</code> , <code>GUID</code> , <code>Date</code> , <code>Timestamp</code> , <code>Decimal</code> , <code>Year</code> , <code>URL</code> , <code>Picklist</code> , <code>Boolean</code>
required	<code>true</code> = must be present for a valid record. <code>false</code> = optional
source	<code>User Input</code> = registry provides this. <code>Inferred (system generated)</code> = auto-generated by CAD Trust. <code>API Generated</code> = set by the API on create/update
picklist	Key into <code>schema.picklists</code> for allowed values (only on picklist fields)
foreignKey	<code>table.field</code> reference (only on FK fields)
aefMapping	Corresponding AEF report field name (only on fields used in Article 6 reporting)
description	Plain-English field description
example	Sample value

Table: project

Core table. Every project record from a registry is stored here. All project-level child tables reference `cad_trust_project_id`.

Column	API Field	Type	Required	Source	Description
<code>cad_trust_project_id</code>	<code>cadTrustProjectId</code>	GUID (PK)	✓	System	Auto-generated identifier
<code>org_uid</code>	<code>orgUid</code>	GUID (FK)	✓	System	Links project registry organization
<code>project_registry_name</code>	<code>projectRegistryName</code>	String	✓	User	Name of the project. e.g. <code>Project 1</code>
<code>project_id</code>	<code>projectId</code>	String	✓	User	Registry's own ID. e.g. <code>3989</code>
<code>project_crediting_program</code>	<code>projectCreditingProgram</code>	String	✗	User	Program the project is part of (if applicable)
<code>project_name</code>	<code>projectName</code>	String	✓	User	Full project name
<code>project_link</code>	<code>projectLink</code>	URL	✓	User	URL to project website
<code>project_description</code>	<code>projectDescription</code>	Text	✗	User	Project description for viewers
<code>project_sector</code>	<code>projectSector</code>	Picklist (Multi)	✓	User	Industry sector. See picklists.projects
<code>project_type</code>	<code>projectType</code>	Picklist (Multi)	✓	User	Project type. See picklists.projects
<code>project_subtype</code>	<code>projectSubtype</code>	Text	✗	User	Free-text subtype
<code>project_status</code>	<code>projectStatus</code>	Picklist	✓	User	Lifecycle status. See picklists.projects
<code>project_status_date</code>	<code>projectStatusDate</code>	Date	✓	User	Date of last status change. Format: <code>DD/MM/YYYY</code>
<code>project_unit_metric</code>	<code>projectUnitMetric</code>	Picklist	✓	User	Unit of measurement. See picklists.projects
<code>cad_trust_reference_project_id</code>	<code>cadTrustReferenceProjectID</code>	String	✗	System	System-generated reference ID. Format: <code>RegId-ProjectId-Last4</code>
<code>cad_trust_program_id</code>	<code>cadTrustProgramID</code>	GUID (FK)	✗	User	Links to <code>program</code> table
<code>created_at</code>	<code>createdAt</code>	Timestamp	✓	API	Auto-generated
<code>updated_at</code>	<code>updatedAt</code>	Timestamp	✓	API	Auto-generated

Table: methodology (NEW v2.0)

Standalone methodology registry. Replaced the free-text `methodology` field on `project` in v1. Supports versioning and multiple methodologies per project via `project_methodology`.

Column	API Field	Type	Required	Description
<code>cad_trust_methodology_id</code>	<code>cadTrustMethodologyId</code>	GUID (PK)	✓	System-generated
<code>methodology_code</code>	<code>methodologyCode</code>	String	✓	Registry code. e.g. <code>GS</code> , <code>VM0007</code>
<code>methodology_name</code>	<code>methodologyName</code>	String	✓	Full methodology name
<code>methodology_type</code>	<code>methodologyType</code>	Picklist	✗	Type of methodology
<code>methodology_version</code>	<code>methodologyVersion</code>	String	✗	Version number. e.g. <code>2.1</code>
<code>methodology_date</code>	<code>methodologyDate</code>	Date	✗	Date methodology is in force
<code>methodology_link</code>	<code>methodologyLink</code>	URL	✗	Link to methodology documentation

Migration from v1: In v1, `project.methodology` was a single string. In v2.0, create a `methodology` record for each unique methodology your registry uses, then link via `project_methodology`.

Table: project_methodology (NEW v2.0)

Junction table. Links projects to methodologies (many-to-many). Also referenced by `issuance` — every issuance must now specify which project methodology it was issued under.

Column	API Field	Type	Required	Description
<code>cad_trust_project_methodology_id</code>	<code>cadTrustProjectMethodologyId</code>	GUID (PK)	✓	System-generated
<code>cad_trust_project_id</code>	<code>cadTrustProjectId</code>	GUID (Composite PK, FK)	✓	→ <code>project</code>
<code>cad_trust_methodology_id</code>	<code>cadTrustMethodologyId</code>	GUID (Composite PK, FK)	✓	→ <code>methodology</code>
<code>project_methodology_date</code>	<code>projectMethodologyDate</code>	Date	✗	Date methodology is in force for this project
<code>project_methodology_description</code>	<code>projectMethodologyDescription</code>	Text	✗	Description of this project-methodology mapping

Key v2.0 change: `issuance.cad_trust_project_methodology_id` is now a required FK. Every issuance must link to a `project_methodology` record.

Table: location (Enhanced v2.0)

Geographic location of a project. One project may have multiple location records.

Column	API Field	Type	Required	Description
<code>cad_trust_location_id</code>	<code>cadTrustLocationId</code>	GUID (PK)	✓	System-generated
<code>location_country</code>	<code>locationCountry</code>	Picklist	✓	ISO 3166 country name. AEF: Table 1 - Party
<code>location_region</code>	<code>locationRegion</code>	Text	✗	Sub-national region or province
<code>location_gis</code>	<code>locationGis</code>	String	✗	GPS coordinates, address, or zone
<code>location_map_type</code>	<code>locationMapType</code>	Picklist	✗	NEW v2.0 GIS file format. See <code>picklists.location_map_type</code>
<code>location_map_file_link</code>	<code>locationMapFileLink</code>	URL	✗	NEW v2.0 URL to GIS file (GeoJSON, Shapefile, GeoPackage, etc.)
<code>cad_trust_project_id</code>	<code>cadTrustProjectId</code>	GUID (FK)	✓	→ <code>project</code>

Table: validation (Restructured v2.0)

Captures the validation event(s) for a project. **In v2.0, validation and verification are now separate tables.** Previously combined in v1.

Column	API Field	Type	Required	Description
<code>cad_trust_validation_id</code>	<code>cadTrustValidationId</code>	GUID (PK)	✓	System-generated
<code>validation_id</code>	<code>validationId</code>	String	✓	Registry's own validation reference. e.g. <code>VCS9 Valid1023CP01</code>
<code>validation_type</code>	<code>validationType</code>	Picklist	✓	See <code>picklists.validation_type</code>
<code>validation_body</code>	<code>validationBody</code>	VVB Picklist	✓	Validation organisation name
<code>validation_date</code>	<code>validationDate</code>	Date	✗	Date validation completed
<code>validation_credit_period_start_date</code>	<code>validationCreditPeriodStartDate</code>	Date	✗	Start of crediting period from validation
<code>validation_credit_period_end_date</code>	<code>validationCreditPeriodEndDate</code>	Date	✗	End of crediting period from validation

Column	API Field	Type	Required	Description
<code>cad_trust_project_id</code>	<code>cadTrustProjectId</code>	GUID (FK)	✓	→ <code>project</code>

Table: verification (*Restructured v2.0*)

Captures periodic verification events. Links to a parent `validation` record and is referenced by `issuance`.

Column	API Field	Type	Required	Description
<code>cad_trust_verification_id</code>	<code>cadTrustVerificationId</code>	GUID (PK)	✓	System-generated
<code>verification_id</code>	<code>verificationId</code>	String	✓	Registry's own verification reference. e.g. <code>VCS9218-V1023</code>
<code>verification_start_date</code>	<code>verificationStartDate</code>	Date	✗	Start of verification period
<code>verification_end_date</code>	<code>verificationEndDate</code>	Date	✗	End of verification period
<code>verification_body</code>	<code>verificationBody</code>	VVB Picklist	✓	Verification organisation name
<code>cad_trust_project_id</code>	<code>cadTrustProjectId</code>	GUID (FK)	✓	→ <code>project</code>
<code>cad_trust_validation_id</code>	<code>cadTrustValidationId</code>	GUID (FK)	✗	→ <code>validation</code> (optional link)

Table: issuance

An issuance event — a batch of units issued under a verification event.

v2.0 change: `cad_trust_project_methodology_id` is now a **required** foreign key. Every issuance must specify the methodology under which it was issued.

Column	API Field	Type	Required	Description
<code>cad_trust_issuance_id</code>	<code>cadTrustIssuanceId</code>	GUID (PK)	✓	System-generated
<code>issuance_id</code>	<code>issuanceId</code>	String	✓	Registry's own issuance reference. e.g. <code>VCS9218-V1023-I212</code>
<code>issuance_date</code>	<code>issuanceDate</code>	Date	✗	Date units were issued
<code>cad_trust_verification_id</code>	<code>cadTrustVerificationId</code>	GUID (FK)	✓	→ <code>verification</code>
<code>cad_trust_project_methodology_id</code>	<code>cadTrustProjectMethodologyId</code>	GUID (FK)	✓	→ <code>project_methodology</code> (NEW required FK in v2.0)
<code>cad_trust_location_id</code>	<code>cadTrustLocationId</code>	GUID (FK)	✓	→ <code>location</code>

Table: unit (*Enhanced v2.0*)

A carbon credit unit block. Four new fields added in v2.0 for retirement detail, beneficiary identity (with LEI support), and ITMOs tracking.

Column	API Field	Type	Required	Notes
<code>cad_trust_unit_id</code>	<code>cadTrustUnitId</code>	GUID (PK)	✓	System-generated
<code>org_uid</code>	<code>orgUid</code>	GUID (FK)	✓	Owning registry organization
<code>unit_serial_id</code>	<code>unitSerialId</code>	String	✓	Registry serial ID. e.g. <code>VCS9819-2014-1233022-1234022</code>
<code>unit_start_block</code>	<code>unitStartBlock</code>	String	✓	First serial in block (inclusive). AEF: Underlying unit block start ID
<code>unit_end_block</code>	<code>unitEndBlock</code>	String	✓	Last serial in block (exclusive). AEF: Underlying unit last block end

Column	API Field	Type	Required	Notes
<code>unit_count</code>	<code>unitCount</code>	Decimal	✓	Total units in this block. AEF: Quantity (t CO2 eq)
<code>unit_type</code>	<code>unitType</code>	Picklist	✓	Mitigation type. See <code>picklists.unit_type</code>
<code>unit_vintage_year</code>	<code>unitVintageYear</code>	Year	✓	Year credits were awarded. AEF: Vintage
<code>unit_status</code>	<code>unitStatus</code>	Picklist	✓	Lifecycle status. See <code>picklists.unit_status</code>
<code>unit_status_reason</code>	<code>unitStatusReason</code>	Text	✓	Reason for current status. Enter <code>N/A</code> if not applicable
<code>unit_status_date</code>	<code>unitStatusDate</code>	Date	✓	Date current status took effect
<code>unit_retirement_detail</code>	<code>unitRetirementDetail</code>	Text	✗	NEW v2.0 Details of retirement (when status = Retired)
<code>unit_retirement_beneficiary</code>	<code>unitRetirementBeneficiary</code>	String	✗	NEW v2.0 Name of retirement beneficiary
<code>unit_retirement_beneficiary_id</code>	<code>unitRetirementBeneficiaryId</code>	String	✗	NEW v2.0 LEI identifier of beneficiary (GLEIF format). e.g. <code>529900T8BM49AURSD055</code>
<code>unit_link</code>	<code>unitLink</code>	URL	✗	URL to unit on issuing registry
<code>unit_metric</code>	<code>unitMetric</code>	Picklist	✓	Unit measurement. AEF: Metric. See <code>picklists.unit_metric</code>
<code>unit_current_owner</code>	<code>unitCurrentOwner</code>	String	✗	Current owner name
<code>unit_itmos_reference_id</code>	<code>unitItmosReferenceId</code>	String	✗	NEW v2.0 ITMO ID per UNFCCC Decision 6/CMA.4 para 17. e.g. <code>CA0004-VU-CH-356-1-3000-2023</code>
<code>cad_trust_issuance_id</code>	<code>cadTrustIssuanceId</code>	GUID (FK)	✓	→ <code>issuance</code>

Table: stakeholder (*NEW v2.0*)

Named participants in a project. Supports multiple stakeholders per project.

Column	API Field	Type	Required	Description
<code>cad_trust_stakeholder_id</code>	<code>cadTrustStakeholderId</code>	GUID (PK)	✓	System-generated
<code>stakeholder_name</code>	<code>stakeholderName</code>	String	✓	Organisation or individual name
<code>stakeholder_type</code>	<code>stakeholderType</code>	Picklist	✓	Role. See <code>picklists.stakeholder_type</code> : Owner, Developer, Consultant
<code>stakeholder_link</code>	<code>stakeholderLink</code>	URL	✗	Company website
<code>cad_trust_project_id</code>	<code>cadTrustProjectId</code>	GUID (FK)	✓	→ <code>project</code>

Table: label (*NEW v2.0*)

Credit eligibility and integrity labels. Labels are not linked directly to units — use `unit_label` (junction table) to associate a label with one or more unit blocks.

Column	API Field	Type	Required	Description
<code>cad_trust_label_id</code>	<code>cadTrustLabelId</code>	GUID (PK)	✓	System-generated
<code>label_name</code>	<code>labelName</code>	Picklist	✓	Label name. e.g. <code>CORSIA</code> , <code>ITMO</code> , <code>SDG</code> , <code>CCP-Approved</code>
<code>label_type</code>	<code>labelType</code>	Picklist	✓	Type of qualification. See <code>picklists.label_type</code>
<code>label_link</code>	<code>labelLink</code>	URL	✗	URL to label details

Column	API Field	Type	Required	Description
<code>label_date</code>	<code>labelDate</code>	Date	✗	Date label was issued or applicable

Table: unit_label (NEW v2.0)

Junction table linking unit blocks to labels. A unit can have multiple labels; a label can apply to multiple units.

Column	API Field	Type	Required	Description
<code>cad_trust_unit_label_id</code>	<code>cadTrustUnitLabelId</code>	GUID (PK)	✓	System-generated
<code>cad_trust_label_id</code>	<code>cadTrustLabelId</code>	GUID (Composite PK, FK)	✓	→ <code>label</code>
<code>cad_trust_unit_id</code>	<code>cadTrustUnitId</code>	GUID (Composite PK, FK)	✓	→ <code>unit</code>
<code>label_unit_date</code>	<code>labelUnitDate</code>	Date (Composite PK)	✓	Date label was applied to this unit block
<code>label_unit_description</code>	<code>label</code>	Text	✗	Additional context on this label-unit application

Table: program (NEW v2.0)

Programme or activity grouping. Projects reference programs via `cad_trust_program_id` on the `project` table.

Column	API Field	Type	Required	Description
<code>cad_trust_program_id</code>	<code>cadTrustProgramId</code>	GUID (PK)	✓	System-generated
<code>program_name</code>	<code>programName</code>	String	✓	Programme name. e.g. <code>Overlapping projects in Country Y</code>
<code>program_registry</code>	<code>programRegistry</code>	String	✓	Registry hosting the program. e.g. <code>Verra</code>
<code>program_registry_id</code>	<code>programRegistryId</code>	String	✓	Registry's own program reference ID. e.g. <code>PROVSC001</code>
<code>program_description</code>	<code>programDescription</code>	Text	✗	Description of programme scope

Tables: co_benefit · estimation · rating

These three tables exist in v1 and are unchanged in v2.0 except for standard `created_at` / `updated_at` timestamps.

co_benefit — SDG co-benefits for a project. `co_benefit_id` is a picklist of all 17 SDGs (see `picklists.co_benefit_id`).

estimation — Expected credit volumes for a future crediting period. Fields: `estimation_start_date`, `estimation_end_date`, `estimation_unit_count` (Decimal, required), `estimation_reference_no` (optional).

rating — Independent assessments. Fields: `rating_type` (CDP / CCQI), `rating_name` (e.g. BeZero), `rating_value`, `rating_link`. All three link back to `project` via `cad_trust_project_id`.

Picklist Values

project_status

`Listed` · `Registered` · `Validated` · `Verified` · `Certified` · `Completed` · `Withdrawn` · `Inactive` · `Rejected`

project_sector (Multi-select)

`Afforestation and reforestation` · `Agriculture` · `Agriculture, forestry and other land use (AFOLU)` · `Carbon capture and storage` · `Chemical industries` · `Construction` · `Energy demand` · `Energy distribution` · `Energy industries (renewable-/ non renewable sources)` · `Fugitive emissions from fuel (solid, oil and gas)` · `Fugitive emissions from the production and consumption of halocarbons and sulfur hexafluoride` · `Livestock and manure management` · `Manufacturing industries` · `Metal production` · `Mining/mineral production` · `Solvent use` · `Transport` · `Urban Development` · `Waste handling and disposal` · `Others` · `Any combination of the above`

project_type (*Multi-select*)

Landfill gas · Hydro · Fossil fuel switch · Wind · Methane avoidance · Biomass Energy · Energy Efficiency households · N2O · Energy Efficiency Industry · Fugitive · Energy Efficiency service · Solar · Cement · Energy Efficiency own generation · Geothermal · Energy Efficiency supply side · Energy distribution · Tidal · Reforestation · Transport · CO2 usage · Coal bed/mine methane · PFCs and SF6 · Afforestation · Agriculture · Biogas · Waste · Any combination of the above

unit_type

Avoidance - nature · Avoidance - technical · Reduction - nature · Reduction - technical · Removal - nature · Removal - technical · Not Determined

unit_status

Issued · Held · Retired · Cancelled · Expired · Buffer · Inactive · Rejected

⚠ v1 → v2 change: **Active** was renamed to **Held**. Any v1 mapping using **Active** must be updated.

unit_metric / project_unit_metric

tCO2e · gCO2eq/kWh · kt (Kiloton)

validation_type

Validation of Project Design Document · Validation of Post Registration Change · Validation of Renewal of Credit Period

location_map_type

Shapefiles · GeoJSON · KML / KMZ · GeoPackage · GML · GPX · GeoTIFF · JPEG 2000 · DXF · TAB · Geodatabases · LiDAR

stakeholder_type

Owner · Developer · Consultant

label_type

Certification · Article 6 - Endorsement · Article 6 - Letter of Qualification · Article 6 - Authorisation · Article 6 - Letter of Approvals

co_benefit_id (*SDGs*)

SDG 1 - No poverty · SDG 2 - Zero hunger · SDG 3 - Good health and well-being · SDG 4 - Quality education · SDG 5 - Gender equality · SDG 6 - Clean water and sanitation · SDG 7 - Affordable and clean energy · SDG 8 - Decent work and economic growth · SDG 9 - Industry, innovation, and infrastructure · SDG 10 - Reduced inequalities · SDG 11 - Sustainable cities and communities · SDG 12 - Responsible consumption and production · SDG 13 - Climate action · SDG 14 - Life below water · SDG 15 - Life on land · SDG 16 - Peace and justice strong institutions · SDG 17 - Partnerships for the goals

rating_type

CDP · CCQI

AEF metric

GHG · Non-GHG

AEF purposes_for_authorization

NDC · OIMP · IMP · OP · NDC and OIMP · NDC and IMP · NDC and OP

AEF action_type

Authorization · Transfer · Acquisition · Use · Cancellation · First Transfer

AEF mitigation_type

Validation Rules

Required fields for data activation

These 30 fields are the minimum a registry must populate for their data to appear in the CAD Trust Data Dashboard. The validator checks these first.

```

project.project_registry_name      project.project_id
project.project_name              project.project_link
project.project_sector            project.project_type
project.project_status            project.project_status_date
project.project_unit_metric

location.location_country

validation.validation_id          validation.validation_type
validation.validation_body

verification.verification_id      verification.verification_body

issuance.issuance_id              issuance.cad_trust_verification_id
issuance.cad_trust_project_methodology_id

unit.unit_serial_id               unit.unit_start_block
unit.unit_end_block               unit.unit_count
unit.unit_type                    unit.unit_vintage_year
unit.unit_status                  unit.unit_status_reason
unit.unit_status_date             unit.unit_metric

methodology.methodology_code      methodology.methodology_name

```

Conditional requirements

Condition	Required field
<code>unit.unit_status = Retired</code>	<code>unit.unit_retirement_detail</code> (recommended)
<code>unit.unit_status = Retired</code>	<code>unit.unit_retirement_beneficiary</code> (recommended)
Unit has Article 6 authorization	<code>unit.unit_itmos_reference_id</code>
Unit carries eligibility labels	<code>unit_label</code> records required
Registry submitting AEF report	All <code>aef_t1</code> through <code>aef_t5</code> tables

Common enum mistakes

✗ Wrong (v1)	✓ Correct (v2.0)	Field
<code>Active</code>	<code>Held</code>	<code>unit_status</code>
<code>Avoidance</code>	<code>Avoidance - nature</code> or <code>Avoidance - technical</code>	<code>unit_type</code>
Single <code>validation</code> record covering both events	Separate <code>validation</code> • <code>verification</code> records	—
Methodology string on <code>project</code>	<code>methodology</code> record + <code>project_methodology</code> junction	—
No <code>cad_trust_project_methodology_id</code> on issuance	Required FK to <code>project_methodology</code>	<code>issuance</code>

Migration Guide — v1 to v2.0

1. methodology

v1: `project.methodology` was a single free-text string.

v2.0: Create a `methodology` record for each unique methodology. Link projects via `project_methodology` junction table.

```
-- For each unique methodology in your v1 data:
INSERT INTO methodology (methodology_code, methodology_name, methodology_version)
VALUES ('VM0007', 'REDD+ Methodology', '1.6');

-- Then link to each project:
INSERT INTO project_methodology (cad_trust_project_id, cad_trust_methodology_id)
VALUES (<project_id>, <new_methodology_id>);
```

2. validation + verification (split)

v1: Single combined record.

v2.0: Create one `validation` record and one `verification` record, optionally linked via `cad_trust_validation_id`.

3. issuance — new required FK

Every `issuance` record now requires `cad_trust_project_methodology_id`. Backfill this by creating `project_methodology` records first, then updating issuance records.

4. unit_type enum values

v1 value	v2.0 mapping
Avoidance	→ Avoidance - nature or Avoidance - technical (registry to determine)
Reduction	→ Reduction - nature or Reduction - technical
Removal	→ Removal - nature or Removal - technical
Active (unit_status)	→ Held

5. Labels (new concept)

If your registry tracks credit eligibility (CORSIA, CCP, ICVCM, etc.), create `label` records and link to units via `unit_label`.

6. program (new structured table)

If your registry previously stored a programme name as a free-text string on `project`, migrate to a `program` record and reference via `cad_trust_program_id` on the project.

Article 6 AEF Tables

Five new tables support UNFCCC Article 6 Agreed Electronic Reporting (AEF). These are only required for registries submitting Article 6 reports.

Table	Purpose
<code>aef_t1_submission</code>	Report submission metadata (party, version, report year, submission date)
<code>aef_t2_authorizations</code>	Authorization records per cooperative approach
<code>aef_t3_actions</code>	ITMO actions (authorization, transfer, acquisition, use, cancellation)
<code>aef_t4_holdings</code>	ITMO holdings records
<code>aef_t5_authorized_entities</code>	Authorized entity registry (corporate entities authorized to use ITMOs)

All AEF tables link back to `project`, `unit`, and `aef_t1_submission`. `aef_t2_authorizations` is the hub — T3, T4, and T5 all reference it via `cad_trust_aef_t2_authorizations_id`.

See the full schema JSON for complete field definitions of all AEF tables.

Download

The full machine-readable schema JSON is available for download and programmatic use:

- **File:** `cadtrust-schema-v2.0.2.json`
- **Format:** JSON Schema (draft/2020-12 compatible)
- **Size:** ~60 KB
- **Contents:** 15 tables · 140 fields · Full picklists · Validation rules · Migration notes

To use in a validator, ETL tool, or AI mapping assistant, load the JSON and reference `schema.tables`, `schema.picklists`, and `schema.validationRules`.

Full JSON Schema

The complete `cadtrust-schema-v2.0.2.json` is embedded below in four sections for readability. Copy all four sections and concatenate them to reconstruct the full file, or download the JSON file directly from this conversation.

Part 1 of 4 — Metadata, Changelog & Picklists

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://climateactiondata.org/schemas/v2.0/cadtrust-data-model.json",
  "title": "CAD Trust Data Model v2.0.2",
  "version": "2.0.2",
  "published": "2026-01-27",
  "source": "https://climateactiondata.org/wp-content/uploads/2026/01/CAD-Trust-Data-Dictionary-v2.0.2-27Jan.xlsx",
  "erd": "https://dbdiagram.io/d/CAD-Trust-V2-0-October-202-68ecbee72e68d21b41249f79",
  "changelog": {
    "v2.0_key_changes": [
      "NEW TABLE: methodology – standalone methodology registry with versioning",
      "NEW TABLE: project_methodology – junction table, supports multiple methodologies per project",
      "NEW TABLE: label – credit eligibility labels (CORSIA, CCP, ITMO, SDG, etc.)",
      "NEW TABLE: unit_label – junction table linking units to labels",
      "NEW TABLE: stakeholder – named project participants (owner, developer, consultant, etc.)",
      "NEW TABLE: program – programme/activity grouping with program_registry_id",
      "NEW TABLES: aef_t1_submission, aef_t2_authorizations, aef_t3_actions, aef_t4_holdings, aef_t5_authorized_entities – Article 6 AEF reporting",
      "RESTRUCTURED: validation split into separate validation and verification tables",
      "NEW FIELDS on unit: unit_retirement_detail, unit_retirement_beneficiary, unit_retirement_beneficiary_id (LEI), unit_itmos_reference_id",
      "NEW FIELD on location: location_map_type (GIS file format picklist), location_map_file_link",
      "ISSUANCE now links to project_methodology via cad_trust_project_methodology_id"
    ],
    "deprecated_fields": [
      "project.methodology (use project_methodology table instead)",
      "project.validationBody (use validation table instead)"
    ]
  },
  "picklists": {
    "project_sector": [
      "Afforestation and reforestation", "Agriculture",
      "Agriculture, forestry and other land use (AFOLU)",

```

```

    "Carbon capture and storage", "Chemical industries", "Construction",
    "Energy demand", "Energy distribution",
    "Energy industries (renewable-/ non renewable sources)",
    "Fugitive emissions from fuel (solid, oil and gas)",
    "Fugitive emissions from the production and consumption of halocarbons and sulfur hexafluoride",
    "Livestock and manure management", "Manufacturing industries",
    "Metal production", "Mining/mineral production", "Solvent use",
    "Transport", "Urban Development", "Waste handling and disposal",
    "Others", "Any combination of the above"
  ],
  "project_type": [
    "Landfill gas", "Hydro", "Fossil fuel switch", "Wind",
    "Methane avoidance", "Biomass Energy", "Energy Efficiency households",
    "N2O", "Energy Efficiency Industry", "Fugitive",
    "Energy Efficiency service", "Solar", "Cement",
    "Energy Efficiency own generation", "Geothermal",
    "Energy Efficiency supply side", "Energy distribution", "Tidal",
    "Reforestation", "Transport", "CO2 usage", "Coal bed/mine methane",
    "PFCs and SF6", "Afforestation", "Agriculture", "Biogas", "Waste",
    "Any combination of the above"
  ],
  "project_status": [
    "Listed", "Registered", "Validated", "Verified", "Certified",
    "Completed", "Withdrawn", "Inactive", "Rejected"
  ],
  "project_unit_metric": ["tCO2e", "gCO2eq/kWh", "kt (Kiloton)"],
  "validation_type": [
    "Validation of Project Design Document",
    "Validation of Post Registration Change",
    "Validation of Renewal of Credit Period"
  ],
  "unit_type": [
    "Avoidance - nature", "Avoidance - technical",
    "Reduction - nature", "Reduction - technical",
    "Removal - nature", "Removal - technical", "Not Determined"
  ],
  "unit_status": [
    "Issued", "Held", "Retired", "Cancelled",
    "Expired", "Buffer", "Inactive", "Rejected"
  ],
  "unit_metric": ["tCO2e", "gCO2eq/kWh", "kt (Kiloton)"],
  "location_map_type": [
    "Shapefiles", "GeoJSON", "KML / KMZ", "GeoPackage",
    "GML", "GPX", "GeoTIFF", "JPEG 2000", "DXF", "TAB",
    "Geodatabases", "LiDAR"
  ],
  "stakeholder_type": ["Owner", "Developer", "Consultant"],
  "label_type": [
    "Certification", "Article 6 - Endorsement",
    "Article 6 - Letter of Qualification",
    "Article 6 - Authorisation", "Article 6 - Letter of Approvals"
  ],
  "co_benefit_id": [
    "SDG 1 - No poverty", "SDG 2 - Zero hunger",
    "SDG 3 - Good health and well-being", "SDG 4 - Quality education",
    "SDG 5 - Gender equality", "SDG 6 - Clean water and sanitation",
    "SDG 7 - Affordable and clean energy",

```

```

    "SDG 8 - Decent work and economic growth",
    "SDG 9 - Industry, innovation, and infrastructure",
    "SDG 10 - Reduced inequalities",
    "SDG 11 - Sustainable cities and communities",
    "SDG 12 - Responsible consumption and production",
    "SDG 13 - Climate action", "SDG 14 - Life below water",
    "SDG 15 - Life on land",
    "SDG 16 - Peace and justice strong institutions",
    "SDG 17 - Partnerships for the goals"
  ],
  "rating_type": ["CDP", "CCQI"],
  "aef_metric": ["GHG", "Non-GHG"],
  "aef_purposes_for_authorization": [
    "NDC", "OIMP", "IMP", "OP", "NDC and OIMP", "NDC and IMP", "NDC and OP"
  ],
  "aef_first_transfer_definition": [
    "First international transfer", "Issuance",
    "Authorization", "Use or cancellation"
  ],
  "aef_t3_action_type": [
    "Authorization", "Transfer", "Acquisition",
    "Use", "Cancellation", "First Transfer"
  ],
  "aef_mitigation_type": ["Emission reductions", "Removals"]
}

```

Part 2 of 4 — Tables: project, methodology, project_methodology, location

```

{
  "tables": {
    "project": {
      "description": "A carbon crediting project registered by an ICP registry. Core table
        - all other project-level tables reference this.",
      "primaryKey": "cad_trust_project_id",
      "fields": {
        "cad_trust_project_id": { "column": "cad_trust_project_id", "apiField": "cadTrustP
          rojectId", "type": "GUID", "required": true, "source": "Inferred (system generated)", "des
            cription": "Unique identifier for projects within the CAD Trust.", "example": "Internal us
              e only" },
        "org_uid": { "column": "org_uid", "apiField": "orgUid", "type": "GUID", "require
          d": true, "source": "Inferred (system generated)", "description": "Unique organization ide
            ntifier linking the project to its owning registry entity." },
        "project_registry_name": { "column": "project_registry_name", "apiField": "project
          RegistryName", "type": "String", "required": true, "source": "User Input", "description":
            "Name of the registry that currently hosts the project.", "example": "Verra" },
        "project_id": { "column": "project_id", "apiField": "projectId", "type": "String",
          "required": true, "source": "User Input", "description": "Identifier used to track the pro
            ject in the current registry.", "example": "3989" },
        "project_crediting_program": { "column": "project_crediting_program", "apiField":
          "projectCreditingProgram", "type": "String", "required": false, "source": "User Input", "d
            escription": "Categorizes the project into a specific program, if applicable.", "example":
              "VCS" },
        "project_name": { "column": "project_name", "apiField": "projectName", "type": "St
          ring", "required": true, "source": "User Input", "description": "The name of the projec
            t.", "example": "CMM Power Generation Project" },
        "project_link": { "column": "project_link", "apiField": "projectLink", "type": "UR

```

```

L", "required": true, "source": "User Input", "description": "URL link to the project on the registry website." },
  "project_description": { "column": "project_description", "apiField": "projectDescription", "type": "Text", "required": false, "source": "User Input", "description": "Description of the project for external viewers." },
  "project_sector": { "column": "project_sector", "apiField": "projectSector", "type": "Picklist, Multi-Select", "required": true, "source": "User Input", "picklist": "project_sector", "description": "Industry sector(s). Multi-select." },
  "project_type": { "column": "project_type", "apiField": "projectType", "type": "Picklist, Multi-Select", "required": true, "source": "User Input", "picklist": "project_type", "description": "Project type(s). Multi-select." },
  "project_subtype": { "column": "project_subtype", "apiField": "projectSubtype", "type": "Text", "required": false, "source": "User Input", "description": "Free-text subtype." },
  "project_status": { "column": "project_status", "apiField": "projectStatus", "type": "Picklist, Single Select", "required": true, "source": "User Input", "picklist": "project_status", "description": "Current lifecycle status." },
  "project_status_date": { "column": "project_status_date", "apiField": "projectStatusDate", "type": "Date", "required": true, "source": "User Input", "description": "Date of last status change. Format: DD/MM/YYYY." },
  "project_unit_metric": { "column": "project_unit_metric", "apiField": "projectUnitMetric", "type": "Picklist, Single Select", "required": true, "source": "User Input", "picklist": "project_unit_metric", "description": "Metric used to measure mitigation outcome s.", "example": "tCO2e" },
  "cad_trust_reference_project_id": { "column": "cad_trust_reference_project_id", "apiField": "cadTrustReferenceProjectID", "type": "String", "required": false, "source": "Inferred (system generated)", "description": "System-generated reference. Format: RegistryName-ProjectId-Last5.", "example": "VER-3989-C0234" },
  "cad_trust_program_id": { "column": "cad_trust_program_id", "apiField": "cadTrustProgramID", "type": "GUID (FK)", "required": false, "source": "User Input", "foreignKey": "program.cad_trust_program_id" },
  "created_at": { "column": "created_at", "apiField": "createdAt", "type": "Timestamp", "required": true, "source": "API Generated" },
  "updated_at": { "column": "updated_at", "apiField": "updatedAt", "type": "Timestamp", "required": true, "source": "API Generated" }
},
"methodology": {
  "description": "NEW in v2.0. Standalone methodology registry with versioning. Replaces free-text methodology field on project.",
  "primaryKey": "cad_trust_methodology_id",
  "fields": {
    "cad_trust_methodology_id": { "column": "cad_trust_methodology_id", "apiField": "cadTrustMethodologyId", "type": "GUID", "required": true, "source": "Inferred (system generated)" },
    "methodology_code": { "column": "methodology_code", "apiField": "methodologyCode", "type": "String", "required": true, "source": "User Input", "description": "Registry code for the methodology.", "example": "GS" },
    "methodology_name": { "column": "methodology_name", "apiField": "methodologyName", "type": "String", "required": true, "source": "User Input", "description": "Full name of the methodology." },
    "methodology_type": { "column": "methodology_type", "apiField": "methodologyType", "type": "Picklist", "required": false, "source": "User Input" },
    "methodology_version": { "column": "methodology_version", "apiField": "methodologyVersion", "type": "String", "required": false, "source": "User Input", "example": "2.1" },
    "methodology_date": { "column": "methodology_date", "apiField": "methodologyDate", "type": "Date", "required": false, "source": "User Input", "description": "Date methodology is in force." },

```

```

    "methodology_link": { "column": "methodology_link", "apiField": "methodologyLink",
    "type": "URL", "required": false, "source": "User Input" },
    "created_at": { "type": "Timestamp", "required": true, "source": "API Generated"
  },
    "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
  },
  },
  "project_methodology": {
    "description": "NEW in v2.0. Junction table: project <-> methodology (many-to-many).
    Required FK on issuance.",
    "primaryKey": "cad_trust_project_methodology_id",
    "compositePK": ["cad_trust_project_id", "cad_trust_methodology_id"],
    "foreignKeys": { "cad_trust_project_id": "project.cad_trust_project_id", "cad_trust_
    methodology_id": "methodology.cad_trust_methodology_id" },
    "fields": {
      "cad_trust_project_methodology_id": { "column": "cad_trust_project_methodology_i
      d", "apiField": "cadTrustProjectMethodologyId", "type": "GUID", "required": true, "sourc
      e": "User Input" },
      "cad_trust_project_id": { "column": "cad_trust_project_id", "apiField": "cadTrustP
      rojectId", "type": "GUID (Composite PK, FK)", "required": true, "foreignKey": "project.cad
      _trust_project_id" },
      "cad_trust_methodology_id": { "column": "cad_trust_methodology_id", "apiField": "c
      adTrustMethodologyId", "type": "GUID (Composite PK, FK)", "required": true, "foreignKey":
      "methodology.cad_trust_methodology_id" },
      "project_methodology_date": { "column": "project_methodology_date", "apiField": "p
      rojectMethodologyDate", "type": "Date", "required": false, "source": "User Input" },
      "project_methodology_description": { "column": "project_methodology_description",
      "apiField": "projectMethodologyDescription", "type": "Text", "required": false, "source":
      "User Input" },
      "created_at": { "type": "Timestamp", "required": true, "source": "API Generated"
    },
      "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
    },
  },
  "location": {
    "description": "Geographic location of a project. Enhanced in v2.0: location_map_type
    and location_map_file_link added.",
    "primaryKey": "cad_trust_location_id",
    "foreignKeys": { "cad_trust_project_id": "project.cad_trust_project_id" },
    "fields": {
      "cad_trust_location_id": { "column": "cad_trust_location_id", "apiField": "cadTrus
      tLocationId", "type": "GUID", "required": true, "source": "Inferred (system generated)" },
      "location_country": { "column": "location_country", "apiField": "locationCountry",
      "type": "Picklist, Single Select", "required": true, "source": "User Input", "picklist":
      "ISO 3166 country names", "aefMapping": "Table 1 - Party", "example": "Indonesia" },
      "location_region": { "column": "location_region", "apiField": "locationRegion", "t
      ype": "Text", "required": false, "source": "User Input", "description": "Sub-national regi
      on or province." },
      "location_gis": { "column": "location_gis", "apiField": "locationGis", "type": "St
      ring", "required": false, "source": "User Input", "description": "GPS coordinates, addres
      s, or zone.", "example": "Lat / Long" },
      "location_map_type": { "column": "location_map_type", "apiField": "locationMapType
      e", "type": "Picklist", "required": false, "source": "User Input", "picklist": "location_m
      ap_type", "description": "NEW v2.0: GIS file format type.", "example": "GeoJSON" },
      "location_map_file_link": { "column": "location_map_file_link", "apiField": "locat
      ionMapFileLink", "type": "URL", "required": false, "source": "User Input", "description":
      "NEW v2.0: URL to GIS file (GeoJSON, Shapefile, GeoPackage, etc.)." },
      "cad_trust_project_id": { "column": "cad_trust_project_id", "apiField": "cadTrustP

```

```

    projectId, "type": "GUID (FK)", "required": true, "foreignKey": "project.cad_trust_project_id" },
    "created_at": { "type": "Timestamp", "required": true, "source": "API Generated"
  },
    "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
  }
}
}
}

```

Part 3 of 4 — Tables: validation, verification, issuance, unit

```

{
  "tables": {
    "validation": {
      "description": "RESTRUCTURED in v2.0. Captures the validation event(s) for a project. Now a separate table from verification.",
      "primaryKey": "cad_trust_validation_id",
      "foreignKeys": { "cad_trust_project_id": "project.cad_trust_project_id" },
      "fields": {
        "cad_trust_validation_id": { "column": "cad_trust_validation_id", "apiField": "cadTrustValidationId", "type": "GUID", "required": true, "source": "Inferred (system generated)" },
        "validation_id": { "column": "validation_id", "apiField": "validationId", "type": "String", "required": true, "source": "User Input", "description": "Registry reference number.", "example": "VCS9218-Valid1023CP01" },
        "validation_type": { "column": "validation_type", "apiField": "validationType", "type": "Picklist, Single Select", "required": true, "source": "User Input", "picklist": "validation_type" },
        "validation_body": { "column": "validation_body", "apiField": "validationBody", "type": "VVB Picklist, Single Select", "required": true, "source": "User Input", "description": "Validation organisation name." },
        "validation_date": { "column": "validation_date", "apiField": "validationDate", "type": "Date", "required": false, "source": "User Input" },
        "validation_credit_period_start_date": { "column": "validation_credit_period_start_date", "apiField": "validationCreditPeriodStartDate", "type": "Date", "required": false, "source": "User Input" },
        "validation_credit_period_end_date": { "column": "validation_credit_period_end_date", "apiField": "validationCreditPeriodEndDate", "type": "Date", "required": false, "source": "User Input" },
        "cad_trust_project_id": { "column": "cad_trust_project_id", "apiField": "cadTrustProjectId", "type": "GUID (FK)", "required": true, "foreignKey": "project.cad_trust_project_id" },
        "created_at": { "type": "Timestamp", "required": true, "source": "API Generated"
      },
        "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
      }
    },
    "verification": {
      "description": "RESTRUCTURED in v2.0. Captures periodic verification events. Links to validation; referenced by issuance.",
      "primaryKey": "cad_trust_verification_id",
      "foreignKeys": { "cad_trust_project_id": "project.cad_trust_project_id", "cad_trust_validation_id": "validation.cad_trust_validation_id" },
      "fields": {
        "cad_trust_verification_id": { "column": "cad_trust_verification_id", "apiField": "cadTrustVerificationId", "type": "GUID", "required": true, "source": "Inferred (system generated)"
      }
    }
  }
}

```

```

nerated)" },
    "verification_id": { "column": "verification_id", "apiField": "verificationId", "type": "String", "required": true, "source": "User Input", "example": "VCS9218-V1023" },
    "verification_start_date": { "column": "verification_start_date", "apiField": "verificationStartDate", "type": "Date", "required": false, "source": "User Input" },
    "verification_end_date": { "column": "verification_end_date", "apiField": "verificationEndDate", "type": "Date", "required": false, "source": "User Input" },
    "verification_body": { "column": "verification_body", "apiField": "verificationBody", "type": "VVB Picklist, Single Select", "required": true, "source": "User Input" },
    "cad_trust_project_id": { "column": "cad_trust_project_id", "apiField": "cadTrustProjectId", "type": "GUID (FK)", "required": true, "foreignKey": "project.cad_trust_project_id" },
    "cad_trust_validation_id": { "column": "cad_trust_validation_id", "apiField": "cadTrustValidationId", "type": "GUID (FK)", "required": false, "foreignKey": "validation.cad_trust_validation_id" },
    "created_at": { "type": "Timestamp", "required": true, "source": "API Generated" },
    "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
  },
  "issuance": {
    "description": "An issuance event. v2.0 change: cad_trust_project_methodology_id is now a REQUIRED FK.",
    "primaryKey": "cad_trust_issuance_id",
    "foreignKeys": { "cad_trust_verification_id": "verification.cad_trust_verification_id", "cad_trust_project_methodology_id": "project_methodology.cad_trust_project_methodology_id", "cad_trust_location_id": "location.cad_trust_location_id" },
    "fields": {
      "cad_trust_issuance_id": { "column": "cad_trust_issuance_id", "apiField": "cadTrustIssuanceId", "type": "GUID", "required": true, "source": "Inferred (system generated)" },
      "issuance_id": { "column": "issuance_id", "apiField": "issuanceId", "type": "String", "required": true, "source": "User Input", "example": "VCS9218-V1023-I212" },
      "issuance_date": { "column": "issuance_date", "apiField": "issuanceDate", "type": "Date", "required": false, "source": "User Input" },
      "cad_trust_verification_id": { "column": "cad_trust_verification_id", "apiField": "cadTrustVerificationId", "type": "GUID (FK)", "required": true, "foreignKey": "verification.cad_trust_verification_id" },
      "cad_trust_project_methodology_id": { "column": "cad_trust_project_methodology_id", "apiField": "cadTrustProjectMethodologyId", "type": "GUID (FK)", "required": true, "foreignKey": "project_methodology.cad_trust_project_methodology_id", "description": "NEW REQUIRED FK in v2.0. Every issuance must link to a project_methodology record." },
      "cad_trust_location_id": { "column": "cad_trust_location_id", "apiField": "cadTrustLocationId", "type": "GUID (FK)", "required": true, "foreignKey": "location.cad_trust_location_id" },
      "created_at": { "type": "Timestamp", "required": true, "source": "API Generated" },
      "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
    }
  },
  "unit": {
    "description": "A carbon credit unit block. Enhanced in v2.0: 4 new fields for retirement detail, beneficiary LEI, and ITMOs tracking.",
    "primaryKey": "cad_trust_unit_id",
    "foreignKeys": { "cad_trust_issuance_id": "issuance.cad_trust_issuance_id" },
    "fields": {
      "cad_trust_unit_id": { "column": "cad_trust_unit_id", "apiField": "cadTrustUnitId", "type": "GUID", "required": true, "source": "Inferred (system generated)" },
      "org_uid": { "column": "org_uid", "apiField": "orgUid", "type": "GUID (FK)", "required": true, "foreignKey": "organization.cad_trust_organization_id" }
    }
  }
}

```

```

    "required": true, "source": "Inferred (system generated)" },
    "unit_serial_id": { "column": "unit_serial_id", "apiField": "unitSerialId", "type": "String", "required": true, "source": "User Input", "example": "VCS9819-2014-1233022-1234022" },
    "unit_start_block": { "column": "unit_start_block", "apiField": "unitStartBlock", "type": "String", "required": true, "source": "User Input", "aefMapping": "Underlying unit block start ID", "example": "1233022" },
    "unit_end_block": { "column": "unit_end_block", "apiField": "unitEndBlock", "type": "String", "required": true, "source": "User Input", "aefMapping": "Underlying unit last block end", "example": "1234023" },
    "unit_count": { "column": "unit_count", "apiField": "unitCount", "type": "Decimal", "required": true, "source": "Inferred", "aefMapping": "Quantity (t CO2 eq)", "example": "1000" },
    "unit_type": { "column": "unit_type", "apiField": "unitType", "type": "Picklist, Single Select", "required": true, "source": "User Input", "picklist": "unit_type", "example": "Removal - nature" },
    "unit_vintage_year": { "column": "unit_vintage_year", "apiField": "unitVintageYear", "type": "Year", "required": true, "source": "User Input", "aefMapping": "Vintage", "example": "2023" },
    "unit_status": { "column": "unit_status", "apiField": "unitStatus", "type": "Picklist, Single Select", "required": true, "source": "User Input", "picklist": "unit_status", "example": "Held" },
    "unit_status_reason": { "column": "unit_status_reason", "apiField": "unitStatusReason", "type": "Text", "required": true, "source": "User Input", "description": "Reason for current status. Enter N/A if not applicable." },
    "unit_status_date": { "column": "unit_status_date", "apiField": "unitStatusDate", "type": "Date", "required": true, "source": "User Input" },
    "unit_retirement_detail": { "column": "unit_retirement_detail", "apiField": "unitRetirementDetail", "type": "Text", "required": false, "source": "User Input", "description": "NEW v2.0. Details of retirement (when status = Retired)."},
    "unit_retirement_beneficiary": { "column": "unit_retirement_beneficiary", "apiField": "unitRetirementBeneficiary", "type": "String", "required": false, "source": "User Input", "description": "NEW v2.0. Name of retirement beneficiary." },
    "unit_retirement_beneficiary_id": { "column": "unit_retirement_beneficiary_id", "apiField": "unitRetirementBeneficiaryId", "type": "String", "required": false, "source": "User Input", "description": "NEW v2.0. LEI (Legal Entity Identifier) from GLEIF.", "example": "529900T8BM49AURSD055" },
    "unit_link": { "column": "unit_link", "apiField": "unitLink", "type": "URL", "required": false, "source": "User Input" },
    "unit_metric": { "column": "unit_metric", "apiField": "unitMetric", "type": "Picklist, Single Select", "required": true, "source": "User Input", "picklist": "unit_metric", "aefMapping": "Metric (default to GHG)", "example": "tCO2e" },
    "unit_current_owner": { "column": "unit_current_owner", "apiField": "unitCurrentOwner", "type": "String", "required": false, "source": "User Input" },
    "unit_itmos_reference_id": { "column": "unit_itmos_reference_id", "apiField": "unitItmosReferenceId", "type": "String", "required": false, "source": "User Input", "description": "NEW v2.0. ITMO ID per UNFCCC Decision 6/CMA.4 para 17.", "example": "CA0004-VU-CH-356-1-3000-2023" },
    "cad_trust_issuance_id": { "column": "cad_trust_issuance_id", "apiField": "cadTrustIssuanceId", "type": "GUID (FK)", "required": true, "foreignKey": "issuance.cad_trust_issuance_id" },
    "created_at": { "type": "Timestamp", "required": true, "source": "API Generated" },
    "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
  }
}

```

```
}
}
```

Part 4 of 4 — Tables: stakeholder, label, unit_label, co_benefit, estimation, rating, program + Validation Rules + Migration Notes

```
{
  "tables": {
    "stakeholder": {
      "description": "NEW in v2.0. Named project participants (owner, developer, consultant).",
      "primaryKey": "cad_trust_stakeholder_id",
      "foreignKeys": { "cad_trust_project_id": "project.cad_trust_project_id" },
      "fields": {
        "cad_trust_stakeholder_id": { "column": "cad_trust_stakeholder_id", "apiField": "cadTrustStakeholderId", "type": "GUID", "required": true, "source": "Inferred (system generated)" },
        "stakeholder_name": { "column": "stakeholder_name", "apiField": "stakeholderName", "type": "String", "required": true, "source": "User Input", "example": "Company Name / Organisation Name" },
        "stakeholder_type": { "column": "stakeholder_type", "apiField": "stakeholderType", "type": "Picklist, Single Select", "required": true, "source": "User Input", "picklist": "stakeholder_type", "example": "Developer" },
        "stakeholder_link": { "column": "stakeholder_link", "apiField": "stakeholderLink", "type": "URL", "required": false, "source": "User Input" },
        "cad_trust_project_id": { "column": "cad_trust_project_id", "apiField": "cadTrustProjectId", "type": "GUID (FK)", "required": true, "foreignKey": "project.cad_trust_project_id" },
        "created_at": { "type": "Timestamp", "required": true, "source": "API Generated" },
        "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
      },
    },
    "label": {
      "description": "NEW in v2.0. Credit eligibility labels (CORSIA, CCP, ITMO, SDG, et c.). Linked to units via unit_label junction table.",
      "primaryKey": "cad_trust_label_id",
      "fields": {
        "cad_trust_label_id": { "column": "cad_trust_label_id", "apiField": "cadTrustLabelId", "type": "GUID", "required": true, "source": "Inferred (system generated)" },
        "label_name": { "column": "label_name", "apiField": "labelName", "type": "Picklist, Single Select", "required": true, "source": "User Input", "example": "CORSIA" },
        "label_type": { "column": "label_type", "apiField": "labelType", "type": "Picklist, Single Select", "required": true, "source": "User Input", "picklist": "label_type", "example": "Certification" },
        "label_link": { "column": "label_link", "apiField": "labelLink", "type": "URL", "required": false, "source": "User Input" },
        "label_date": { "column": "label_date", "apiField": "labelDate", "type": "Date", "required": false, "source": "User Input" },
        "created_at": { "type": "Timestamp", "required": true, "source": "API Generated" },
        "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
      },
    },
    "unit_label": {
      "description": "NEW in v2.0. Junction table: unit <-> label (many-to-many).",
    }
  }
}
```

```

    "primaryKey": "cad_trust_unit_label_id",
    "compositePK": ["cad_trust_label_id", "cad_trust_unit_id"],
    "foreignKeys": { "cad_trust_label_id": "label.cad_trust_label_id", "cad_trust_unit_id": "unit.cad_trust_unit_id" },
    "fields": {
      "cad_trust_unit_label_id": { "column": "cad_trust_unit_label_id", "apiField": "cadTrustUnitLabelId", "type": "GUID", "required": true, "source": "API Generated" },
      "cad_trust_label_id": { "column": "cad_trust_label_id", "apiField": "cadTrustLabelId", "type": "GUID (Composite PK, FK)", "required": true, "foreignKey": "label.cad_trust_label_id" },
      "cad_trust_unit_id": { "column": "cad_trust_unit_id", "apiField": "cadTrustUnitId", "type": "GUID (Composite PK, FK)", "required": true, "foreignKey": "unit.cad_trust_unit_id" },
      "label_unit_date": { "column": "label_unit_date", "apiField": "labelUnitDate", "type": "Date (Composite PK)", "required": true, "source": "User Input", "description": "Date label was applied to this unit block." },
      "label_unit_description": { "column": "label_unit_description", "apiField": "labelUnitDescription", "type": "Text", "required": false, "source": "User Input" },
      "created_at": { "type": "Timestamp", "required": true, "source": "API Generated" },
      "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
    },
    "co_benefit": {
      "description": "SDG co-benefits associated with a project.",
      "primaryKey": "cad_trust_co_benefit_id",
      "foreignKeys": { "cad_trust_project_id": "project.cad_trust_project_id" },
      "fields": {
        "cad_trust_co_benefit_id": { "column": "cad_trust_co_benefit_id", "apiField": "cadTrustCobenefitId", "type": "GUID", "required": true, "source": "Inferred (system generated)" },
        "co_benefit_id": { "column": "co_benefit_id", "apiField": "cobenefit", "type": "Picklist, Single Select", "required": true, "source": "User Input", "picklist": "co_benefit_id", "example": "SDG 13 - Climate action" },
        "cad_trust_project_id": { "column": "cad_trust_project_id", "apiField": "cadTrustProjectId", "type": "GUID (FK)", "required": true, "foreignKey": "project.cad_trust_project_id" },
        "created_at": { "type": "Timestamp", "required": true, "source": "API Generated" },
        "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
      },
      "estimation": {
        "description": "Expected credit volumes for a future crediting period.",
        "primaryKey": "cad_trust_estimation_id",
        "foreignKeys": { "cad_trust_project_id": "project.cad_trust_project_id" },
        "fields": {
          "cad_trust_estimation_id": { "column": "cad_trust_estimation_id", "apiField": "cadTrustEstimationId", "type": "GUID", "required": true, "source": "Inferred (system generated)" },
          "estimation_start_date": { "column": "estimation_start_date", "apiField": "estimationStartDate", "type": "Date", "required": true, "source": "User Input" },
          "estimation_end_date": { "column": "estimation_end_date", "apiField": "estimationEndDate", "type": "Date", "required": true, "source": "User Input" },
          "estimation_unit_count": { "column": "estimation_unit_count", "apiField": "estimationUnitCount", "type": "Decimal", "required": true, "source": "User Input", "example": "1000" },
          "estimation_reference_no": { "column": "estimation_reference_no", "apiField": "estimationReferenceNo", "type": "Text", "required": true, "source": "User Input", "example": "1000" }
        }
      }
    }
  }
}

```

```

    imationReferenceNo", "type": "String", "required": false, "source": "User Input", "example": "E12332321" },
    "cad_trust_project_id": { "column": "cad_trust_project_id", "apiField": "cadTrustProjectId", "type": "GUID (FK)", "required": true, "foreignKey": "project.cad_trust_project_id" },
    "created_at": { "type": "Timestamp", "required": true, "source": "API Generated"
  },
  "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
},
"rating": {
  "description": "Independent project ratings (BeZero, CDP, CCQI, etc.).",
  "primaryKey": "cad_trust_rating_id",
  "foreignKeys": { "cad_trust_project_id": "project.cad_trust_project_id" },
  "fields": {
    "cad_trust_rating_id": { "column": "cad_trust_rating_id", "apiField": "cadTrustRatingId", "type": "GUID", "required": true, "source": "Inferred (system generated)" },
    "rating_type": { "column": "rating_type", "apiField": "ratingType", "type": "Picklist, Single Select", "required": true, "source": "User Input", "picklist": "rating_type", "example": "CDP" },
    "rating_name": { "column": "rating_name", "apiField": "ratingName", "type": "String", "required": true, "source": "User Input", "example": "BeZero" },
    "rating_value": { "column": "rating_value", "apiField": "ratingValue", "type": "String", "required": true, "source": "User Input", "example": "5" },
    "rating_link": { "column": "rating_link", "apiField": "ratingLink", "type": "URL", "required": false, "source": "User Input" },
    "cad_trust_project_id": { "column": "cad_trust_project_id", "apiField": "cadTrustProjectId", "type": "GUID (FK)", "required": true, "foreignKey": "project.cad_trust_project_id" },
    "created_at": { "type": "Timestamp", "required": true, "source": "API Generated"
  },
  "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
},
"program": {
  "description": "NEW in v2.0. Programme or activity grouping. Projects reference via cad_trust_program_id on project table.",
  "primaryKey": "cad_trust_program_id",
  "fields": {
    "cad_trust_program_id": { "column": "cad_trust_program_id", "apiField": "cadTrustProgramId", "type": "GUID", "required": true, "source": "Inferred (system generated)" },
    "program_name": { "column": "program_name", "apiField": "programName", "type": "String", "required": true, "source": "User Input", "example": "Overlapping projects in County Y" },
    "program_registry": { "column": "program_registry", "apiField": "programRegistry", "type": "String", "required": true, "source": "User Input", "example": "Verra" },
    "program_registry_id": { "column": "program_registry_id", "apiField": "programRegistryId", "type": "String", "required": true, "source": "User Input", "example": "PROVSC001" },
    "program_description": { "column": "program_description", "apiField": "programDescription", "type": "Text", "required": false, "source": "User Input" },
    "created_at": { "type": "Timestamp", "required": true, "source": "API Generated"
  },
  "updated_at": { "type": "Timestamp", "required": true, "source": "API Generated" }
}
},
"validationRules": {

```

```

    "required_for_data_activation": [
        "project.project_registry_name", "project.project_id", "project.project_name",
        "project.project_link", "project.project_sector", "project.project_type",
        "project.project_status", "project.project_status_date", "project.project_unit_metric",
        "location.location_country",
        "validation.validation_id", "validation.validation_type", "validation.validation_body",
        "verification.validation_id", "verification.validation_body",
        "issuance.issuance_id", "issuance.cad_trust_verification_id", "issuance.cad_trust_project_methodology_id",
        "unit.unit_serial_id", "unit.unit_start_block", "unit.unit_end_block", "unit.unit_count",
        "unit.unit_type", "unit.unit_vintage_year", "unit.unit_status", "unit.unit_status_reason",
        "unit.unit_status_date", "unit.unit_metric",
        "methodology.methodology_code", "methodology.methodology_name"
    ],
    "conditional_requirements": {
        "unit.unit_retirement_detail": "Recommended when unit.unit_status = Retired",
        "unit.unit_retirement_beneficiary": "Recommended when unit.unit_status = Retired",
        "unit.unit_itmos_reference_id": "Required for Article 6 ITMO units",
        "unit_label_records": "Required if unit carries eligibility labels (CORSIA, CCP, etc.)",
        "aef_tables": "Required only for registries submitting Article 6 AEF reports"
    },
    "enum_notes": [
        "All picklist values are case-sensitive. Match exactly.",
        "project_sector and project_type are Multi-Select – pass as comma-separated string or array.",
        "unit_status 'Active' was renamed to 'Held' in v2.0. Update all v1 mappings.",
        "validation and verification are now separate tables in v2.0. Split combined v1 records.",
        "unit_type 'Avoidance' is now split into 'Avoidance - nature' and 'Avoidance - technical'.",
        "issuance now requires cad_trust_project_methodology_id – create project_methodology records first."
    ],
    "date_format": "DD/MM/YYYY for user-input dates. Timestamps are auto-generated by the API."
},
"migrationNotes": {
    "from_v1": {
        "methodology": "v1 stored methodology as a string field on project. v2.0 requires a separate methodology record + project_methodology junction. Split each unique methodology into its own record.",
        "validation_verification": "v1 had a combined validation/verification concept. v2.0 splits them. Create one validation record and one verification record per audit event, linked via cad_trust_validation_id.",
        "unit_type": "v1 'Avoidance' -> 'Avoidance - nature' or 'Avoidance - technical'. v1 'Reduction' -> 'Reduction - nature' or 'Reduction - technical'. v1 'Removal' -> 'Removal - nature' or 'Removal - technical'.",
        "unit_status": "v1 'Active' -> 'Held' in v2.0.",
        "new_required_fk_on_issuance": "issuance now requires cad_trust_project_methodology_id. Create project_methodology records first, then backfill issuance records.",
        "labels": "v1 had no label concept. Map any eligibility flags (CORSIA, CCP, etc.) to label + unit_label records.",
        "program": "v1 project.program was a free-text string. Migrate unique programme name

```

```
s into program table records and reference via cad_trust_program_id on project."  
    }  
  }  
}
```